

checks is odd. Set a parity bit to 0 if the total number of 1's in the positions that it checks is even. This means even parity is ensured in the code word.

We illustrate with an example. Say, the original given byte of data is 10011010. As per Hamming code, the code word will be: 1 0 0 1 1 0 1 0, where ' ' locations are for the parity for each parity bit.

- Location-1 check bit is 0, as created by even parity rule of data bit locations 1, 3, 5, 7, 9, 11.
- Location-2 check bit is obtained by even parity rule of data bit locations 2, 3, 6, 7, 10, 11 as 1.
- Location-4 check bit is obtained from data locations of 4, 5, 6, 7, 12 as 1.
- Location-8 check bit is obtained from data locations of 8, 9, 10, 11, 12 as 0.
- Thus the code word becomes: 011100101010.

Suppose the received code is

011100101110. Here the error location is highlighted in bold. The receiver calculates parity bits from the received data, and compares calculated parity bits with received parity bits to find out which bit is in order to correct it. The method is to verify all the incorrect parity bits by the comparison stated. In the example, parity bits 2 and 8 are incorrect.

It is now $2 + 8 = 10$, and that bit position 10 is the location of the incorrect bit. We complement the tenth bit to get back the correct code word.

Repetition code

Coding Theory is the study of how to add redundancy or additional bits to the original given data so as to use them to detect and correct errors induced by the communication channel. Here, the transmitter sends the data bit several times, an odd number of times in fact. In general, if each bit is repeated $2K + 1$ (K is a positive integer) times, the code can tolerate up to K errors.

In repetition code, bit to be transmitted is transmitted more than once. In triple repetition code, '0' and '1' are coded, respectively, as '000' and '111'. At the receiver, majority rule is applied to decide about the bit. If the three bits received were not identical, an error occurred. If the channel is nearly clean, most likely only one bit will change in each triple. Therefore 001, 010 and 100 each correspond to a 0 bit, while 110, 101 and 011 correspond to a 1 bit.

Such codes cannot correct all errors. For example, if the channel introduces two bits error and the receiver gets '001', the system detects the error, but concludes that the original bit was 0, which is incorrect. If we increase the number of times we duplicate each bit to four, we can detect all two-bit errors but can't correct; at five, we can correct all two-bit errors, but not detect all three-bit errors. •

The author is a regular contributor to EYF

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